



VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN

(Course Code: ECE 1003)

Module-1:Lecture-5

Number Systems

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BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Number Systems

SIGNED BINARY NUMBERS

✓ For signed numbers

- **Positive** numbers => + **sign**
- **Negative** numbers => - **sign**

0 1010
↑ ↑
Sign Magnitude

✓ However, in **digital system** i.e. computers

- **Positive** numbers => 0
 - **Negative** numbers => 1
- } The MSB bit represents the sign bit

✓ **For example:** if a binary number is given as

- 1) 01010 → 10 (unsigned number)
+10 (signed number)
- 2) 11010 → 26 (unsigned number)
-10 (signed number)

01010

Represents +ve sign

11010

Represents -ve sign

SIGNED BINARY NUMBERS

Signed magnitude form

- ✓ The number consists of magnitude bits and a sign bit
- ✓ Used in ordinary arithmetic
- ✓ It is simple

Drawbacks:

- ✓ Hardware limitations

Decimal	Binary	Decimal	Binary
+0	0 000	-0	1 000
+1	0 001	-1	1 001
+2	0 010	-2	1 010
+3	0 011	-3	1 011
+4	0 100	-4	1 100
+5	0 101	-5	1 101
+6	0 110	-6	1 110
+7	0 111	-7	1 111

SIGNED BINARY NUMBERS

- ✓ The signed binary number makes the arithmetic difficult because the sum of the negative and positive number is not zero.
- ✓ For example:

$$+1 = 0001$$

$$\underline{-1 = 1001}$$

$$1010 \Rightarrow -2 \text{ (not correct)}$$

- ✓ Hence, to overcome the above difficulty, the **complement methods** can be used **to represent the -ve numbers**.
- ✓ In 1's complement or 2's complement of any **+ve number**, the representation is **similar to signed magnitude representation**. However, the **-ve numbers** should be represented in **complement form**.

SIGNED BINARY NUMBERS

1's Complement Representation

- ✓ To find negative of 1's complement number take the 1's complement of that positive number.

$$\begin{array}{ccc} & \textcolor{brown}{0} & \underline{1100}_2 = +12_{10} \\ & \nearrow & \nwarrow \\ \text{Sign bit} & & \text{Magnitude} \end{array}$$

$$\begin{array}{ccc} & \textcolor{brown}{1} & \underline{0011}_2 = -12_{10} \\ & \nearrow & \nwarrow \\ \text{Sign bit} & & \text{Magnitude} \end{array}$$

2's Complement Representation

- ✓ To find negative of 2's complement number take the 1's complement of the +ve number and add 1.

$$\begin{array}{ccc} & \textcolor{brown}{0} & \underline{1100}_2 = +12_{10} \\ & \nearrow & \nwarrow \\ \text{Sign bit} & & \text{Magnitude} \end{array}$$

$$\begin{array}{ccc} & \textcolor{brown}{1} & \underline{0100}_2 = -12_{10} \\ & \nearrow & \nwarrow \\ \text{Sign bit} & & \text{Magnitude} \end{array}$$

SIGNED BINARY NUMBERS

✓ Hence, a –ve number can be represented in 3 ways

- **Signed Magnitude Representation**
- **1's Complement Representation**
- **2's Complement Representation**

✓ For example :

– **7** Representation

Signed Magnitude Representation

1 111

1's Complement Representation

1 000

2's Complement Representation

1 001

id	b ₃	b ₂	b ₁	b ₀	Signed	One's	Two's
0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1
2	0	0	1	0	2	2	2
3	0	0	1	1	3	3	3
4	0	1	0	0	4	4	4
5	0	1	0	1	5	5	5
6	0	1	1	0	6	6	6
7	0	1	1	1	7	7	7
8	1	0	0	0	-0	-7	-8
9	1	0	0	1	-1	-6	-7
10	1	0	1	0	-2	-5	-6
11	1	0	1	1	-3	-4	-5
12	1	1	0	0	-4	-3	-4
13	1	1	0	1	-5	-2	-3
14	1	1	1	0	-6	-1	-2
15	1	1	1	1	-7	-0	-1

SIGNED BINARY NUMBERS

- ✓ Machines that use 2's complement arithmetic can represent integers in the range

$$-(2^{n-1}) \leq N \leq (2^{n-1}) - 1$$

where n is the number of bits available for representing N . Note that $(2^{n-1}) - 1 = (011..11)_2$ and $-(2^{n-1}) = (100..00)_2$

- ✓ The **arithmetic operations** are generally performed using **4-bit, 8-bit, 16-bit or 32-bit**, so on.
- ✓ For example, the **8-bit representation** of -9 in 3 different form

signed-magnitude representation: 10001001

signed-1's-complement representation: 11110110

signed-2's-complement representation: 11110111

Note: (8-bit Representation)

+7 $\rightarrow$ **00000111** $\rightarrow$ **1's Complement** $\rightarrow$ **11111000** $\rightarrow$ **-7**
 $\rightarrow$ **2's Complement** $\rightarrow$ **11111001** $\rightarrow$ **-7**

SIGNED BINARY NUMBERS

Express the decimal number -39 as an 8-bit number in the sign-magnitude, 1's complement, and 2's complement forms.

First, write the 8-bit number for $+39$.

00100111

In the *sign-magnitude form*, -39 is produced by changing the sign bit to a 1 and leaving the magnitude bits as they are. The number is

10100111

In the *1's complement form*, -39 is produced by taking the 1's complement of $+39$ (00100111).

11011000

In the *2's complement form*, -39 is produced by taking the 2's complement of $+39$ (00100111) as follows:

11011000	1's complement
$\pm$ _____ 1	
11011001	2's complement

SIGNED BINARY NUMBERS

Determine the decimal values of the signed binary numbers expressed in 1's complement:

(a) 00010111 (b) 11101000

(a) The bits and their powers-of-two weights for the positive number are as follows:

$$\begin{array}{rcccccccc} -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{array}$$

Summing the weights where there are 1s,

$$16 + 4 + 2 + 1 = +23$$

(b) The bits and their powers-of-two weights for the negative number are as follows. Notice that the negative sign bit has a weight of -2^7 or -128 .

$$\begin{array}{rcccccccc} -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{array}$$

Summing the weights where there are 1s,

$$-128 + 64 + 32 + 8 = -24$$

Adding 1 to the result, the final decimal number is

$$-24 + 1 = -23$$

SIGNED BINARY NUMBERS

Determine the decimal values of the signed binary numbers expressed in 2's complement:

(a) 01010110 (b) 10101010

(a) The bits and their powers-of-two weights for the positive number are as follows:

$$\begin{array}{cccccccc} -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \end{array}$$

Summing the weights where there are 1s,

$$64 + 16 + 4 + 2 = +86$$

(b) The bits and their powers-of-two weights for the negative number are as follows. Notice that the negative sign bit has a weight of $-2^7 = -128$.

$$\begin{array}{cccccccc} -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{array}$$

Summing the weights where there are 1s,

$$-128 + 32 + 8 + 2 = -86$$

SIGNED BINARY NUMBERS

Example-1: compute $+12_{10} + (+1)_{10}$ using both 1's and 2's complement method

- $(+12)_{10} = +(1100)_2 = 01100_2$ in **1's complement** and 01100_2 in **2's complement**
- $(+1)_{10} = +(0001)_2 = 00001_2$ in **1's complement** and 00001_2 in **2's complement**

1's complement Method

```

      0 1 1 0 0
    + 0 0 0 0 1
    -----
Final Result  0 1 1 0 1

```

2's complement Method

```

      0 1 1 0 0
    + 0 0 0 0 1
    -----
Final Result  0 1 1 0 1

```

$$(01101)_2 = +13_{10}$$

SIGNED BINARY NUMBERS

Example-2: compute $+12_{10} + (-1)_{10}$ using both 1's and 2's complement method

- $(+12)_{10} = +(1100)_2 = 01100_2$ in **1's complement** and 01100_2 in **2's complement**
- $(-1)_{10} = -(0001)_2 = 11110_2$ in **1's complement** and 11111_2 in **2's complement**

1's complement Method

$$\begin{array}{r} \\ + \\ \hline \\ \text{End Around Carry } \textcircled{1} \\ \\ \hline \text{Final Result} \end{array}$$

2's complement Method

$$\begin{array}{r} \\ + \\ \hline \\ \text{Discard End Carry } \textcircled{1} \\ \\ \hline \text{Final Result} \end{array}$$

$$(01101)_2 = +11_{10}$$

SIGNED BINARY NUMBERS

Example-3: compute $+5_{10} + (-12)_{10}$ using both 1's and 2's complement method

- $(+5)_{10} = +(0101)_2 = 00101_2$ in **1's complement** and 00101_2 in **2's complement**
- $(-12)_{10} = -(1100)_2 = 10011_2$ in **1's complement** and 10100_2 in **2's complement**

1's complement Method

$$\begin{array}{r} \text{MSB} = -\text{ve} \leftarrow \begin{array}{r} 0 \ 0 \ 1 \ 0 \ 1 \\ + \ 1 \ 0 \ 0 \ 1 \ 1 \\ \hline 1 \ 1 \ 0 \ 0 \ 0 \end{array} \end{array}$$

Final Result = $-(1\text{'s complement of } (11000)_2)$
= $-(00111)_2$

2's complement Method

$$\begin{array}{r} \text{MSB} = -\text{ve} \leftarrow \begin{array}{r} 0 \ 0 \ 1 \ 0 \ 1 \\ + \ 1 \ 0 \ 1 \ 0 \ 0 \\ \hline 1 \ 1 \ 0 \ 0 \ 1 \end{array} \end{array}$$

Final Result = $-(2\text{'s complement of } (11001)_2)$
= $-(00111)_2$

$$-(00111)_2 = -7_{10}$$

SIGNED BINARY NUMBERS

Example-4: compute $-5_{10} + (12)_{10}$ using both 1's and 2's complement method

- $(-5)_{10} = -(0101)_2 = \textcolor{red}{1}1010_2$ in **1's complement** and $\textcolor{red}{1}1011_2$ in **2's complement**
- $(12)_{10} = +(1100)_2 = \textcolor{red}{0}1100_2$ in **1's complement** and $\textcolor{red}{0}1100_2$ in **2's complement**

1's complement Method

$$\begin{array}{r}
 1\ 1\ 0\ 1\ 0 \\
 +\ 0\ 1\ 1\ 0\ 0 \\
 \hline
 1\ 0\ 0\ 1\ 1\ 0 \\
 \hline
 \text{End Around Carry} \rightarrow 1 \\
 \hline
 \text{Final Result} = 0\ 0\ 1\ 1\ 1
 \end{array}$$

2's complement Method

$$\begin{array}{r}
 1\ 1\ 0\ 1\ 1 \\
 +\ 0\ 1\ 1\ 0\ 0 \\
 \hline
 1\ 0\ 0\ 1\ 1\ 1 \\
 \hline
 \end{array}$$

Discard End Carry ←

Final Result = $0\ 0\ 1\ 1\ 1$

$$(00111)_2 = +7_{10}$$

SIGNED BINARY NUMBERS

Example-5: compute $-5_{10} + (-9)_{10}$ using both 1's and 2's complement method

- $(-5)_{10} = -(0101)_2 = 11010_2$ in **1's complement** and 11011_2 in **2's complement**
- $(-9)_{10} = -(1001)_2 = 10110_2$ in **1's complement** and 10111_2 in **2's complement**

1's complement Method

MSB = -ve

$$\begin{array}{r} 1\ 1\ 0\ 1\ 0 \\ +\ 1\ 0\ 1\ 1\ 0 \\ \hline 1\ 1\ 0\ 0\ 0\ 0 \\ \text{End Around Carry} \rightarrow 1 \\ \hline 1\ 0\ 0\ 0\ 1 \end{array}$$

Final Result = $-(1\text{'s complement of } (10001)_2)$
 $= -(01110)_2$

2's complement Method

MSB = -ve

$$\begin{array}{r} 1\ 1\ 0\ 1\ 1 \\ +\ 1\ 0\ 1\ 1\ 1 \\ \hline 1\ 1\ 0\ 0\ 1\ 0 \end{array}$$

Discard End Carry

Final Result = $-(2\text{'s complement of } (10010)_2)$
 $= -(01110)_2$
 $-(01110)_2 = -14_{10}$

SIGNED BINARY NUMBERS

Why 8-bit representation?

- ✓ Consider the following signed bit addition example

$$\begin{array}{r} + 9 \quad \quad 0 \ 1001 \\ + 11 \quad + 0 \ 1011 \\ \hline + 20 \quad 1 \ 0100 \end{array} = -4 \text{ (incorrect)}$$

- ✓ In the above example, the 5th bit is considered as signed bit. However, the **carry bit replaces the signed bit** that can be observed from the result obtained which makes the results incorrect.
- ✓ To overcome this issue, it is recommended to use **8-bit representation** for addition. But the number should be in the range $-128 < N < 127$ and the sum should not exceed 255.

$$\begin{array}{r} + 9 \quad \quad 0 \ 0001001 \\ + 11 \quad + 0 \ 0001011 \\ \hline + 20 \quad \quad 0 \ 0010100 \\ \quad \quad \quad + \quad 20 \end{array}$$

SIGNED BINARY NUMBERS

Example-1: compute $21 + (-7)$ using both 1's and 2's complement method using 8-bit

1's complement Method

MSB = +ve ←

	0	0	0	1	0	1	0	1
+	1	1	1	1	1	0	0	0

End Around	1	0	0	0	0	1	1	0
Carry								1

	0	0	0	0	1	1	1	0

(1's Comp. of -7)

Final Result = $(00001110)_2$

2's complement Method

MSB = +ve ←

	0	0	0	1	0	1	0	1
+	1	1	1	1	1	0	0	1

Discard End	1	0	0	0	0	1	1	1
Carry								0

	0	0	0	0	1	1	1	0

(2's Comp. of -7)

Final Result = $(00001110)_2$

$$(00001110)_2 = 14_{10}$$

SIGNED BINARY NUMBERS

Example-2: compute $17 + (-18)$ using both 1's and 2's complement method using 8-bit

1's complement Method

$$\begin{array}{r} \text{MSB} = - \text{ve} \leftarrow \begin{array}{cccccccc} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ + & 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & \text{(1's Comp. of -18)} \\ \hline 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \end{array} \end{array}$$

$$\begin{aligned} \text{Result} &= - (1\text{'s complement of } (11111110)_2) \\ &= -(00000001)_2 \end{aligned}$$

2's complement Method

$$\begin{array}{r} \text{MSB} = - \text{ve} \leftarrow \begin{array}{cccccccc} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ + & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & \text{(2's Comp. of -18)} \\ \hline 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{array} \end{array}$$

$$\begin{aligned} \text{Result} &= - (2\text{'s complement of } (11111111)_2) \\ &= -(00000001)_2 \end{aligned}$$

$$-(00000001)_2 = -1_{10}$$

SIGNED BINARY NUMBERS

Example-3: compute $-44 + 13$ using both 1's and 2's complement method using 8-bit

1's complement Method

$$\begin{array}{r} \text{MSB} = -\text{ve} \quad \leftarrow 1 \ 1 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1 \text{ (1's Comp. of -44)} \\ + \quad 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ 1 \\ \hline 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \end{array}$$

$$\begin{aligned} \text{Result} &= - (1\text{'s complement of } (11100000)_2) \\ &= -(00011111)_2 \end{aligned}$$

2's complement Method

$$\begin{array}{r} \text{MSB} = -\text{ve} \quad \leftarrow 1 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \text{ (2's Comp. of -44)} \\ + \quad 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ 1 \\ \hline 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1 \end{array}$$

$$\begin{aligned} \text{Result} &= - (2\text{'s complement of } (11100001)_2) \\ &= -(00011111)_2 \end{aligned}$$

$$-(00011111)_2 = -31_{10}$$

SIGNED BINARY NUMBERS

Example-4: compute $-44 + 68$ using both 1's and 2's complement method using 8-bit

1's complement Method

MSB = +ve ←

	1	1	0	1	0	0	1	1	(1's Comp. of -7)
+	0	1	0	0	0	1	0	0	

End Around	1	0	0	0	1	0	1	1	1
Carry									1

	0	0	0	1	1	0	0	0	

Final Result = $(00011000)_2$

2's complement Method

MSB = +ve ←

	1	1	0	1	0	1	0	0	(2's Comp. of -7)
+	0	1	0	0	0	1	0	0	

Discard End	1	0	0	0	1	1	0	0	0
Carry									

	0	0	0	1	1	0	0	0	

Final Result = $(00011000)_2$

$$(00011000)_2 = +24_{10}$$

SIGNED BINARY NUMBERS

Example-5: compute $-32 + (-53)$ using both 1's and 2's complement method using 8-bit

1's complement Method

MSB = -ve

	1	1	0	1	1	1	1	(1's Comp. of -32)
+	1	1	0	0	1	0	1	(1's Comp. of -53)

End Around	1	1	0	1	0	1	0	1
Carry								1

	1	0	1	0	1	0	1	0

$$\begin{aligned}\text{Result} &= - (1's \text{ complement of } (10101010)_2 \\ &= -(01010101)_2\end{aligned}$$

2's complement Method

MSB = -ve

	1	1	1	0	0	0	0	(2's Comp. of -32)
+	1	1	0	0	1	0	1	(2's Comp. of -53)

Discard End	1	1	0	1	0	1	0	1
Carry								1

	1	0	1	0	1	0	1	1

$$\begin{aligned}\text{Result} &= - (2's \text{ complement of } (10101011)_2 \\ &= -(01010101)_2\end{aligned}$$

$$-(01010101)_2 = -85_{10}$$

SIGNED BINARY NUMBERS

Both numbers positive:

$$\begin{array}{r} 00000111 \quad 7 \\ + 00000100 \quad + 4 \\ \hline 00001011 \quad 11 \end{array}$$

The sum is positive and is therefore in true (uncomplemented) binary.

Positive number with magnitude larger than negative number:

$$\begin{array}{r} 00001111 \quad 15 \\ + 1111010 \quad + -6 \\ \hline \end{array}$$

Discard carry $\longrightarrow$ 1 00001001 9

The final carry bit is discarded. The sum is positive and therefore in true (uncomplemented) binary.

Negative number with magnitude larger than positive number:

$$\begin{array}{r} 00010000 \quad 16 \\ + 11101000 \quad + -24 \\ \hline 11111000 \quad -8 \end{array}$$

The sum is negative and therefore in 2's complement form.

Both numbers negative:

$$\begin{array}{r} 11111011 \quad -5 \\ + 11110111 \quad + -9 \\ \hline \end{array}$$

Discard carry $\longrightarrow$ 1 1110010 -14

SIGNED BINARY NUMBERS

Perform each of the following subtractions of the signed numbers:

(a) $00001000 - 00000011$ (b) $00001100 - 11110111$

(c) $11100111 - 00010011$ (d) $10001000 - 11100010$

Like in other examples, the equivalent decimal subtractions are given for reference.

(a) In this case, $8 - 3 = 8 + (-3) = 5$.

	00001000	Minuend (+8)
	<u>+ 1111101</u>	2's complement of subtrahend (-3)
Discard carry →	1 0000101	Difference (+5)

(b) In this case, $12 - (-9) = 12 + 9 = 21$.

	00001100	Minuend (+12)
	<u>+ 00001001</u>	2's complement of subtrahend (+9)
	00010101	Difference (+21)

(c) In this case, $-25 - (+19) = -25 + (-19) = -44$.

	11100111	Minuend (-25)
	<u>+ 11101101</u>	2's complement of subtrahend (-19)
Discard carry →	1 11010100	Difference (-44)

(d) In this case, $-120 - (-30) = -120 + 30 = -90$.

	10001000	Minuend (-120)
	<u>+ 00011110</u>	2's complement of subtrahend (+30)
	10100110	Difference (-90)

ASSIGNMENT QUESTIONS

Question: compute

a) $-12 + (-33)$ using both 1's and 2's complement method using 8-bit

a) $98 + (-9)$ using both 1's and 2's complement method using 8-bit

Thank You